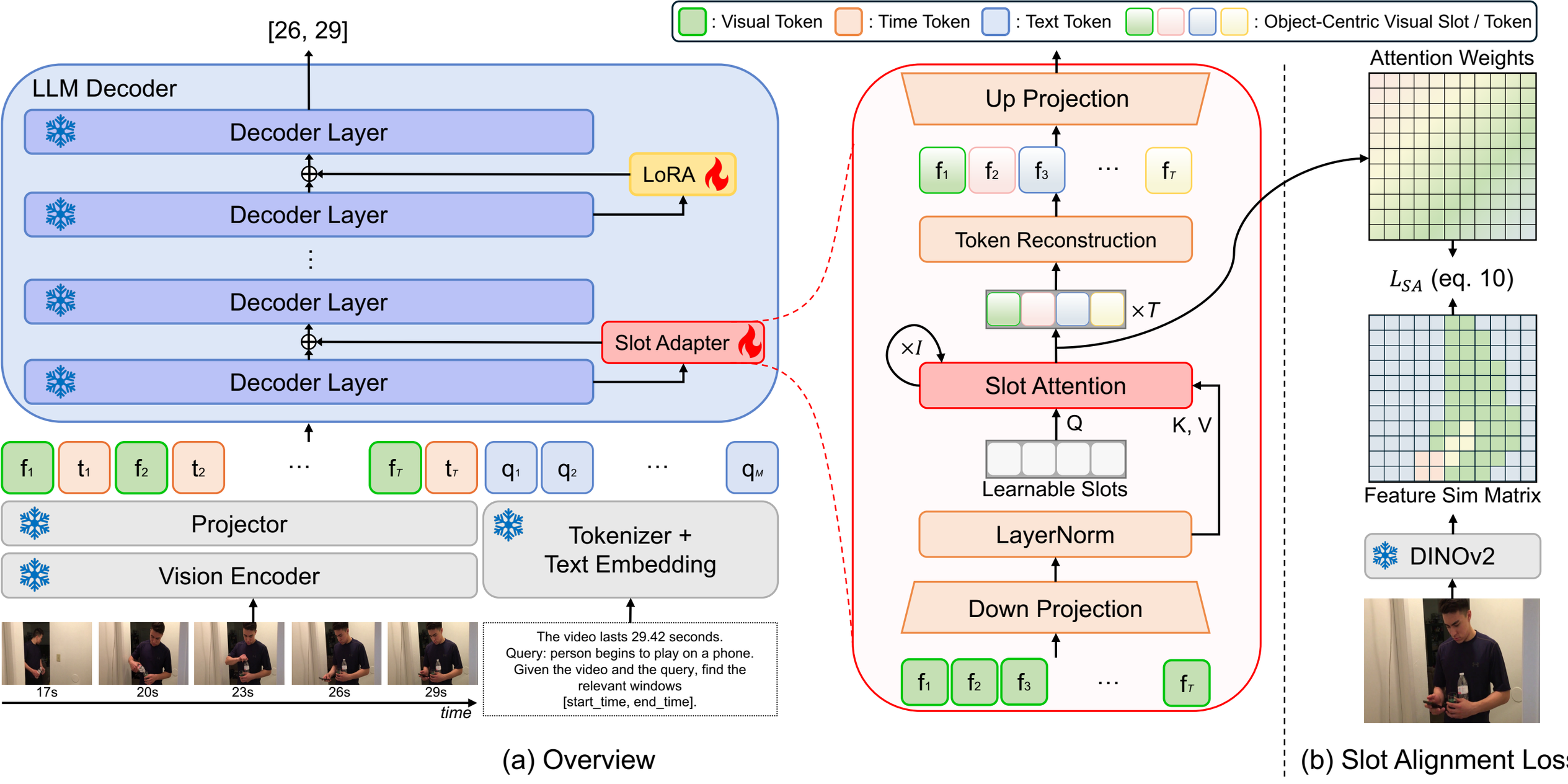

: Visual Token
: Time Token
: Text Token
: Object-Centric Visual Slot / Token
[26, 29]
LLM Decoder
Decoder Layer
LoRA
Decoder Layer
Decoder Layer
Slot Adapter
Decoder Layer
f1 t1 f2 t2 ... fT tT q1 q2 ... qM
Projector
Vision Encoder
Tokenizer + Text Embedding
17s
20s
23s
26s
29s
time
The video lasts 29.42 seconds.
Query: person begins to play on a phone.
Given the video and the query, find the relevant windows [start_time, end_time].
(a) Overview
Up Projection
f1 f2 f3 ... fT
Token Reconstruction
×T
×I
Slot Attention
Q
K, V
Learnable Slots
LayerNorm
Down Projection
f1 f2 f3 ... fT
Attention Weights
L_{SA} (eq. 10)
Feature Sim Matrix
DINOv2
(b) Slot Alignment Loss